

Is There a Better, More Affordable Alternative to AIMD? Validating Foundational MLIPs on Solid-State Sodium Ionic Conductors

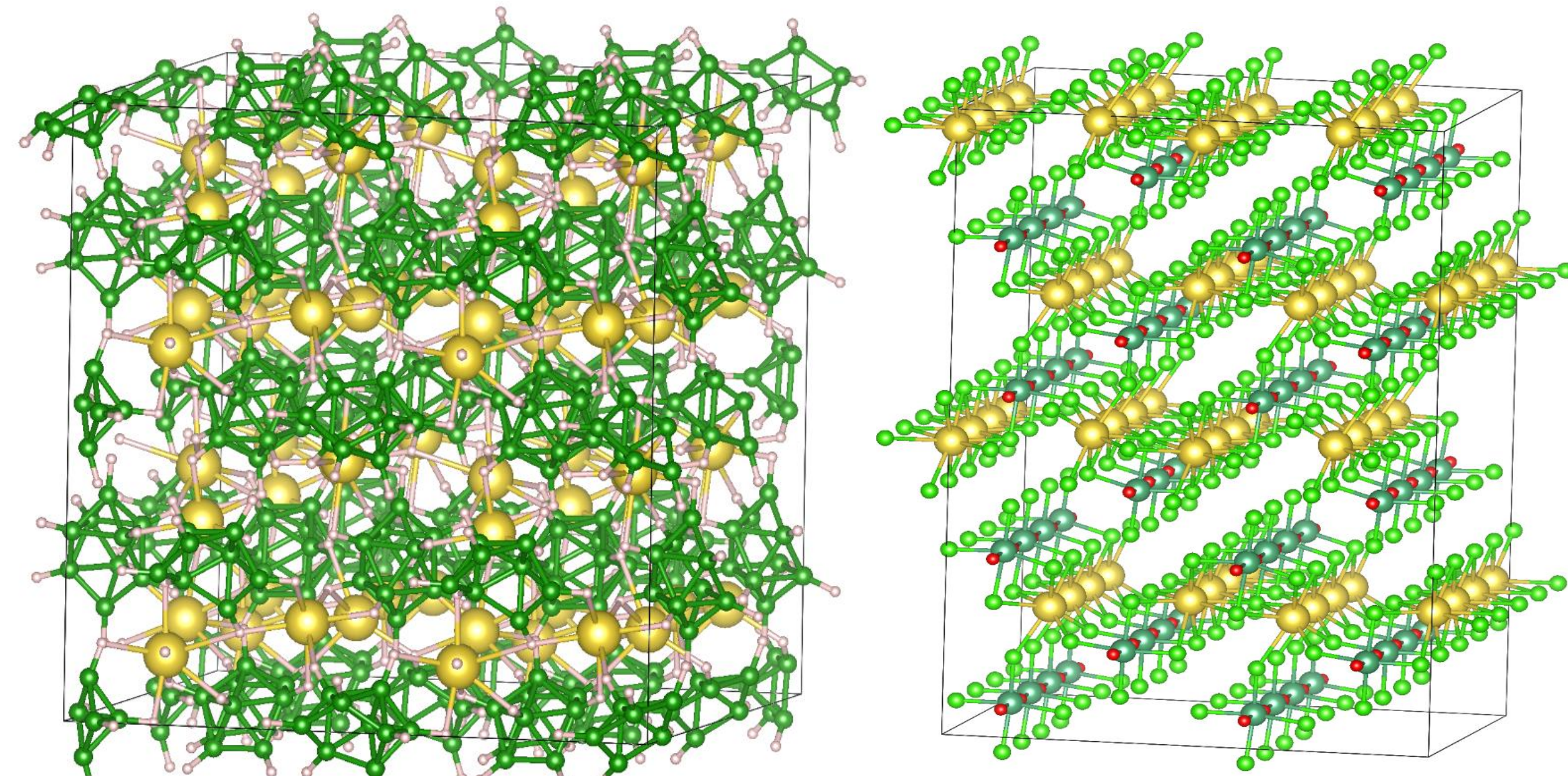
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Motivation

- Solid-state sodium ionic conductors are a promising class of materials for next-generation battery electrolytes, and understanding their ion transport is critical for design
- Ab Initio Molecular Dynamics (AIMD) is the gold standard for simulating these systems but costs hundreds of dollars and weeks of compute time per system
- Foundation Machine Learning Interatomic Potentials like MACE-MP-0b2 promise equivalent accuracy in minutes on a GPU based on our runs with zero training required
- Hydrogen dynamics is hard to model correctly due to its light mass and high vibrational frequencies, making hydridic systems a rigorous test case

What We Studied

- NBH: pure $\text{Na}_2\text{B}_{10}\text{H}_{10}$, pure $\text{Na}_2\text{B}_{12}\text{H}_{12}$, and mixed phase — simulated at 700K and 800K
- NNOC: experimentally synthesized and computationally derived lowest-energy structure — simulated at 500K
- MACE-MP-0b2 validated against AIMD (PBE) using identical MD settings and fully independent runs



Crystal structures of mixed $\text{Na}_2\text{B}_{10}\text{H}_{10}/\text{Na}_2\text{B}_{12}\text{H}_{12}$ (left) and NaNbOCl_4 (right)

Earth Mover's Distance Table

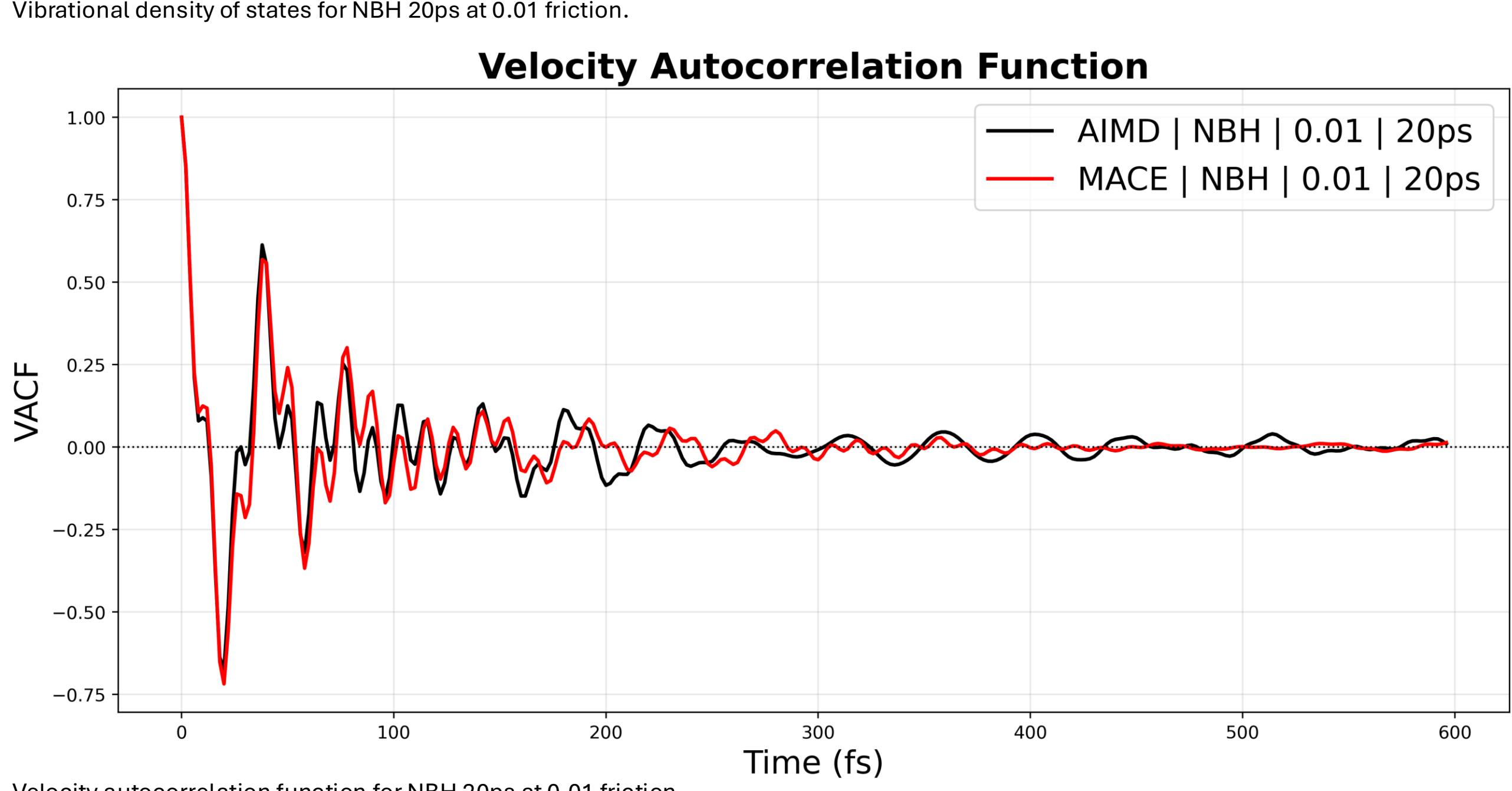
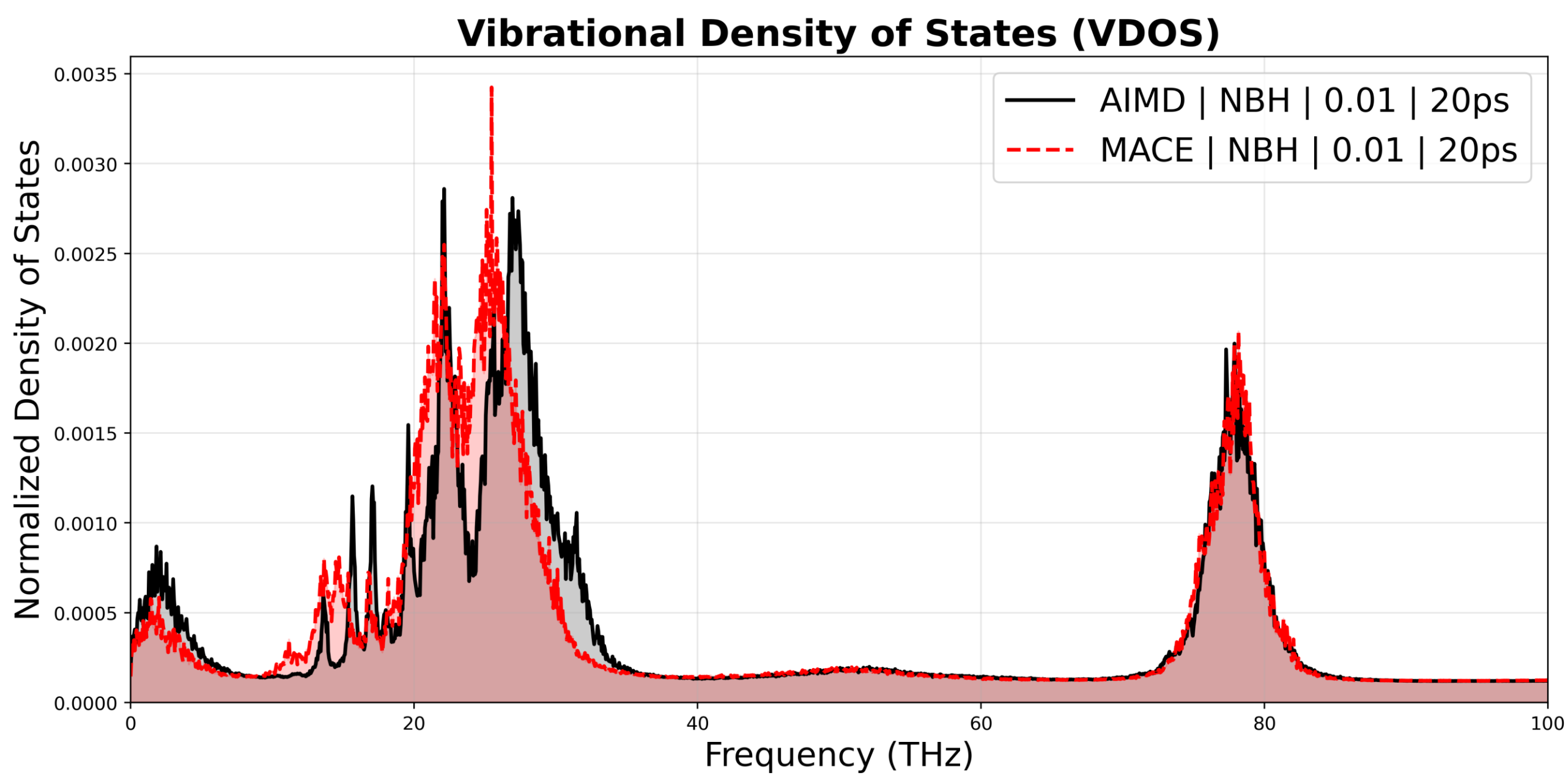
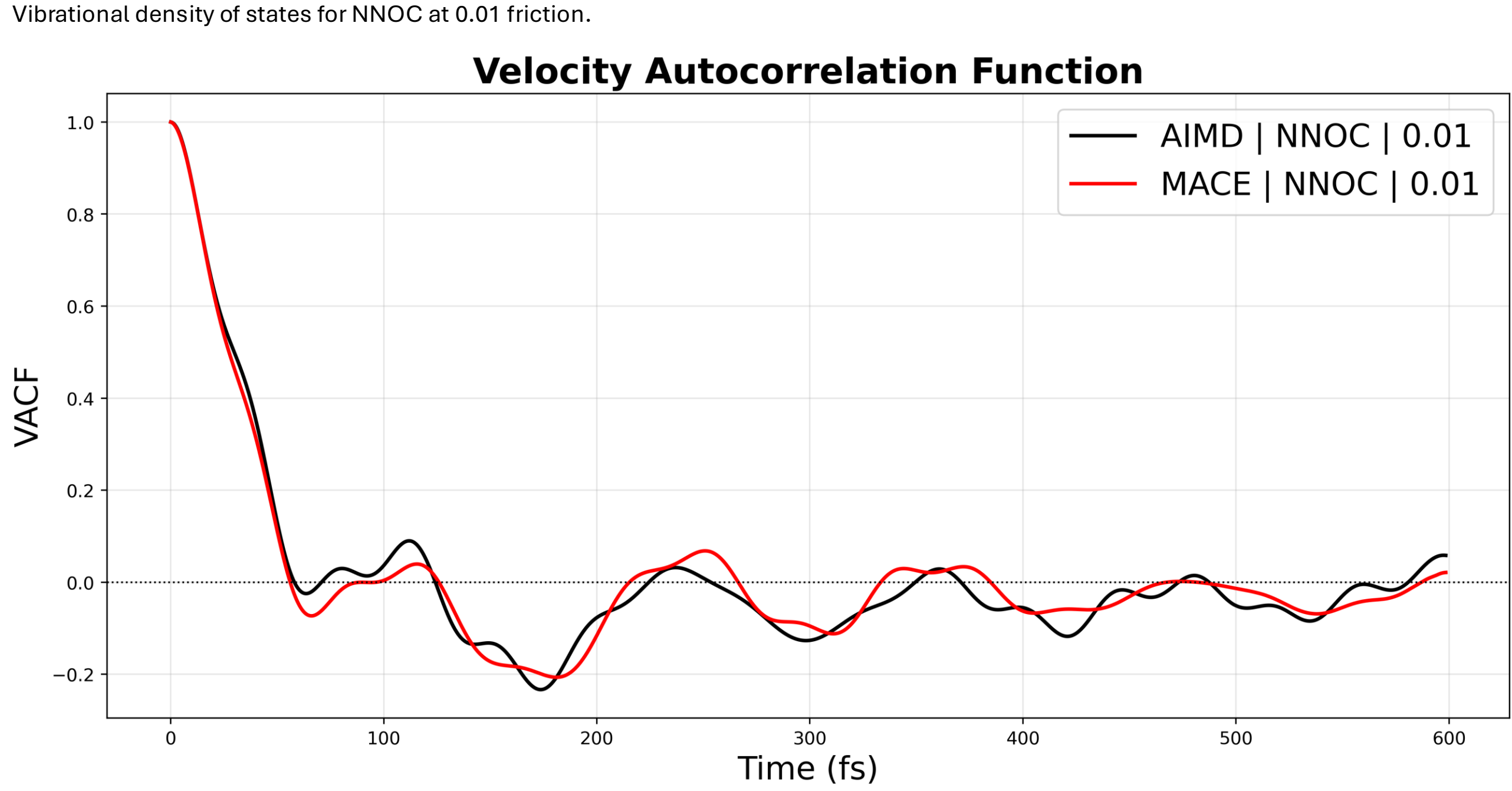
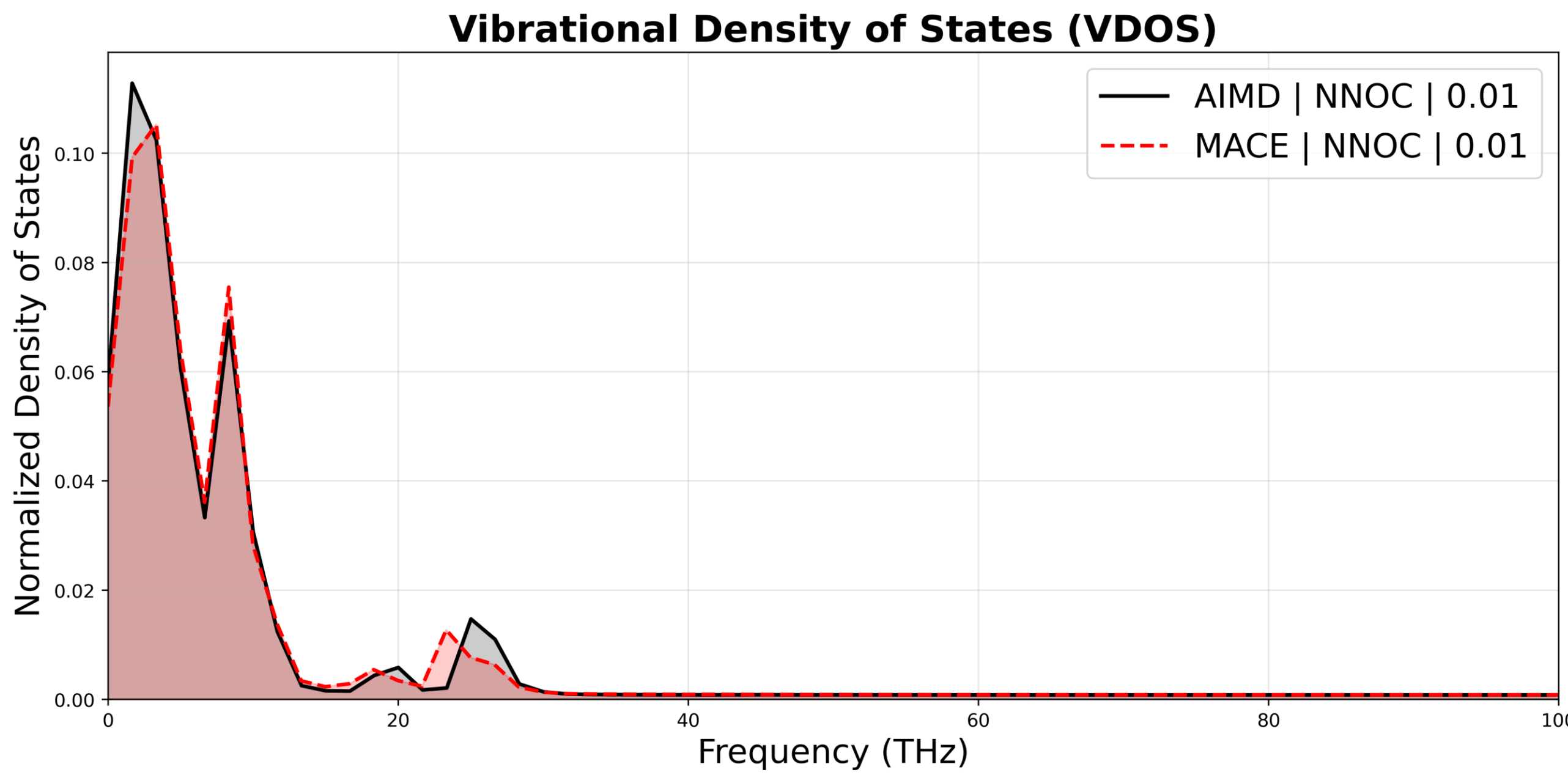
Model / Run	Epot (eV/atom)	Ekin (eV/atom)	RDF (Å)	ADF (°)	VDOS (THz)	Epot raw (eV)	Ekin raw (eV)	Cv Error (eV/K/atom)
NBH 700K 20ps 0.01	0.00656	1.28	0.0042	0.26	0.64	0.63	122.62	0.00038
NBH 700K 20ps 0.25	0.00444	1.28	0.0043	0.31	2.42	0.43	122.40	0.00038
NBH 700K 1ps 0.01	0.01594	162.33	0.0116	0.11	131.43	12.24	124670.38	0.00006
NBH 700K 1ps 0.25	0.0046	162.32	0.0124	0.39	126.65	3.54	124658.02	0.00006
NNOC 500K exp 0.01	0.012	0.016	0.0080	0.81	0.75	5.46	7.15	0.00014
NNOC 500K exp 0.25	0.0043	0.00088	0.010	0.50	11.28	1.93	0.39	0.00008

Earth Mover's Distance (EMD) quantifies the difference between MACE and AIMD distributions. Lower values indicate better agreement. Energy observables reported both per atom and as system totals to match figure axes.

Future Work

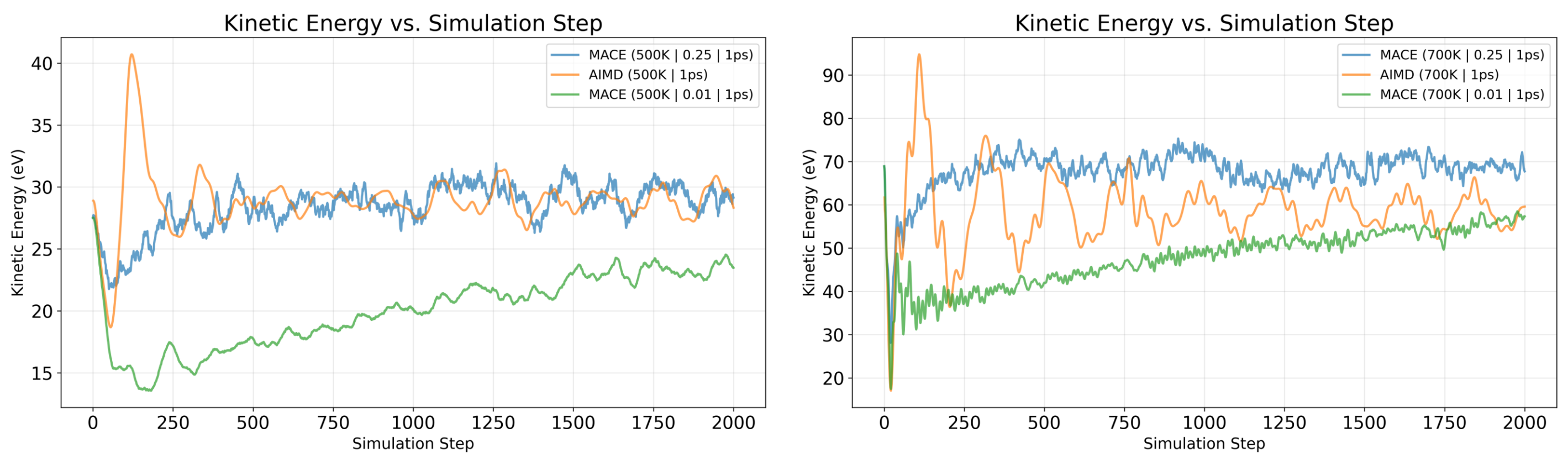
- Continue development and refinement of the EMD-based validation pipeline
- Investigate remaining discrepancies in NBH systems and extend to additional compositions

Vibrational Properties: Friction and Simulation Length Effects



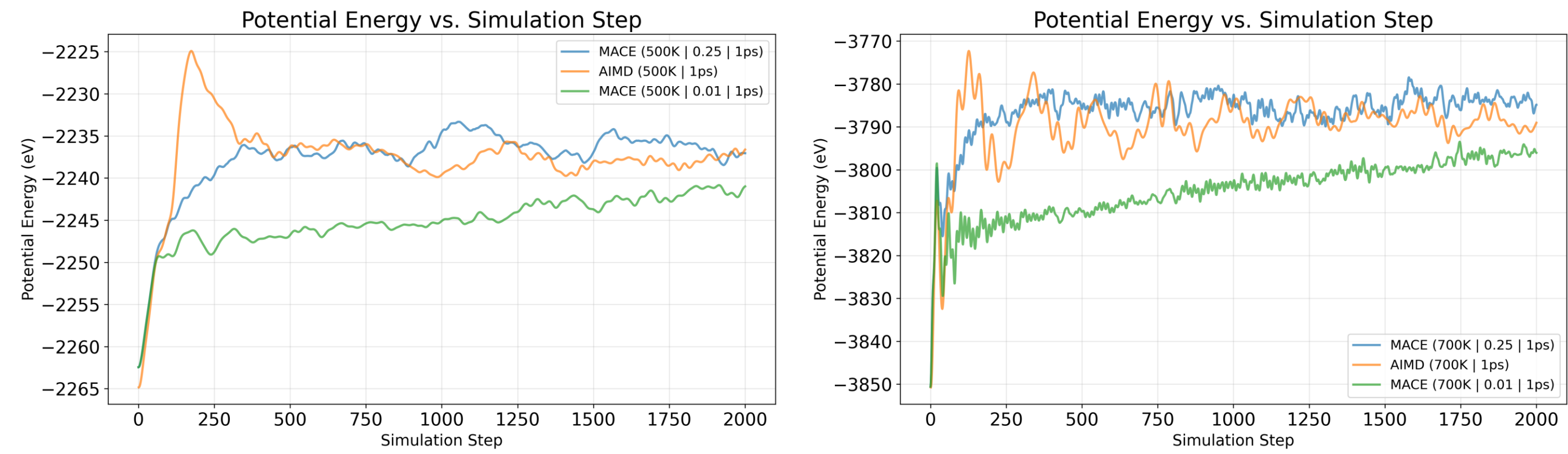
Thermodynamic Properties

Kinetic Energy plots



Kinetic energy vs. simulation step for NNOC (left) and NBH (right) at 0.01 and 0.25 friction. MACE accurately reproduces NNOC kinetic energy but shows significant error in hydrogen-containing NBH.

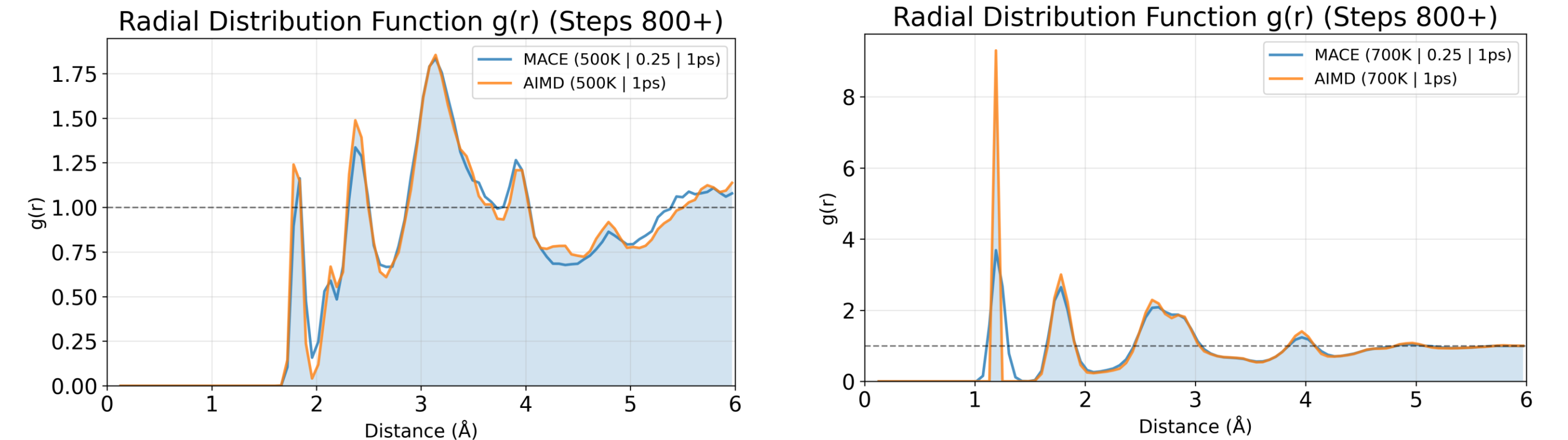
Potential Energy plots



Potential energy vs. simulation step for NNOC (left) and NBH (right). MACE reproduces potential energy well across both systems regardless of friction.

Structural Properties

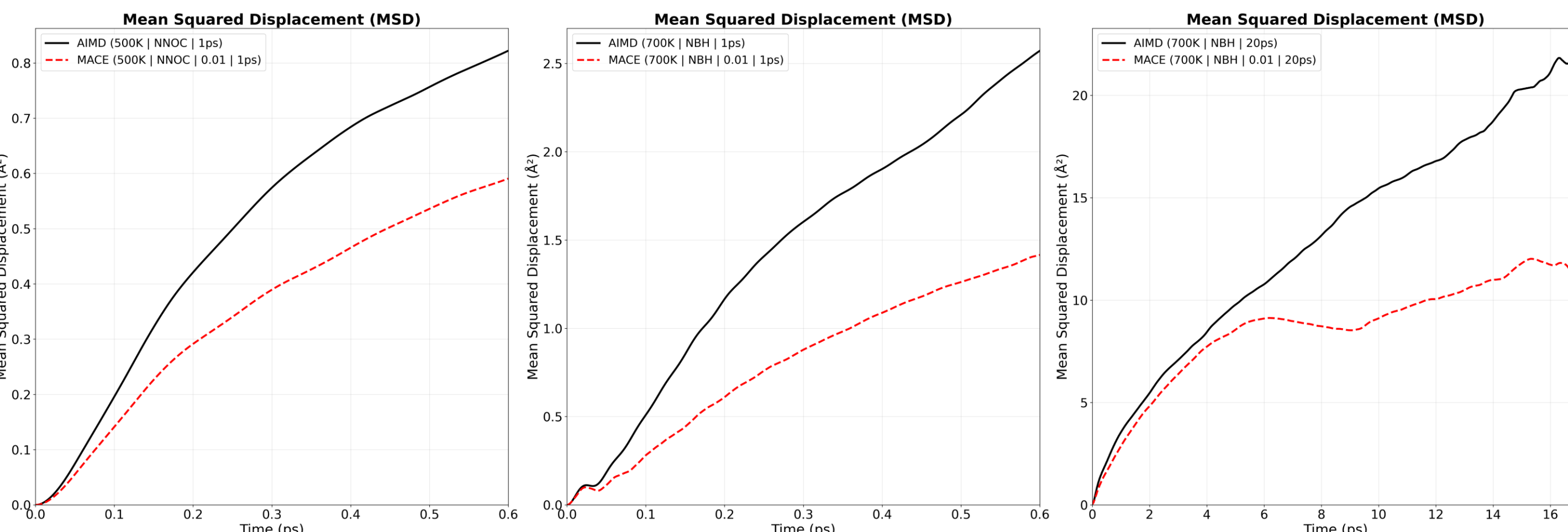
Radial Distribution Function



Radial distribution function $g(r)$ for NBH (left) and NNOC (right) at 0.25 friction. MACE reproduces the general structural features of both systems with reasonable agreement.

Dynamic Properties

Mean Squared Displacement



Mean squared displacement for NNOC (left), NBH 1ps (middle), and NBH 20ps (right) at 0.01 friction.

Conclusion

This study evaluated MACE-MP-0b2 against AIMD using a reusable EMD-based validation pipeline. Hydrogen appears to play a significant role in MLIP accuracy, with NBH showing substantially higher kinetic energy error than NNOC. Thermostat friction critically affects which observables are reliable.